6 Register Mapping

This chapter gives an overview of the complete register set. Some of the registers bundling a number of single bits are detailed in extra tables. The functional practical application of the settings is detailed in dedicated chapters.

Note
- All registers become reset to 0 upon power up, unless otherwise noted.
- Add 0x80 to the address **Addr** for write accesses!

NOTATION OF HEXADECIMAL AND BINARY NUMBERS	
0x	precedes a hexadecimal number, e.g. 0x04
%	precedes a multi-bit binary number, e.g. %100

NOTATION OF R/W FIELD	
R	Read only
W	Write only
R/W	Read- and writable register
R+WC	Clear by writing "1" bit

OVERVIEW REGISTER MAPPING

REGISTER	DESCRIPTION
General Configuration Registers	These registers contain - global configuration - global status flags - interface configuration - and I/O signal configuration
Ramp Generator Motion Control Register Set	This register set offers registers for - choosing a ramp mode - choosing velocities - homing - acceleration and deceleration - target positioning - reference switch and StallGuard2 event configuration - ramp and reference switch status
Velocity Dependent Driver Feature Control Register Set	This register set offers registers for - driver current control - setting thresholds for CoolStep operation - setting thresholds for different chopper modes - setting thresholds for DcStep operation
Encoder Register Set	The encoder register set offers all registers needed for proper ABN encoder operation.
Motor Driver Register Set	This register set offers registers for - setting / reading out microstep table and counter - chopper and driver configuration - CoolStep and StallGuard2 configuration - DcStep configuration - reading out StallGuard2 values and driver error flags

6.1 General Configuration Registers

GENERAL CONFIGURATION REGISTERS (0X00...0X0F)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	
RW	0x00	18	*GCONF*	**Bit**	***GCONF* – Global configuration flags**
				0	*recalibrate* 1: Zero crossing recalibration during driver disable (via DRV_ENN or via *TOFF* setting)
				1	*faststandstill* Timeout for step execution until standstill detection: 1: Short time: 2^18 clocks 0: Normal time: 2^20 clocks
				2	*en_pwm_mode* 1: StealthChop voltage PWM mode enabled (depending on velocity thresholds). Switch from off to on state while in stand-still and at IHOLD= nominal IRUN current, only.
				3	*multistep_filt* 1: Enable step input filtering for StealthChop optimization with external step source (default=1)
				4	*shaft* 1: Inverse motor direction
				5	*diag0_error* (only with SD_MODE=1) 1: Enable DIAG0 active on driver errors: Over temperature (*ot*), short to GND (*s2g*) DIAG0 always shows the reset-status, i.e., is active low during reset condition.
				6	*diag0_otpw* (only with SD_MODE=1) 1: Enable DIAG0 active on driver over temperature prewarning (*otpw*)
				7	*diag0_stall* (with SD_MODE=1) 1: Enable DIAG0 active on motor stall (set *TCOOLTHRS* before using this feature) *diag0_step* (with SD_MODE=0) 0: DIAG0 outputs interrupt signal 1: Enable DIAG0 as STEP output (half frequency, dual edge triggered) for external STEP/DIR driver
				8	*diag1_stall* (with SD_MODE=1) 1: Enable DIAG1 active on motor stall (set *TCOOLTHRS* before using this feature) *diag1_dir* (with SD_MODE=0) 0: DIAG1 outputs position compare signal 1: Enable DIAG1 as DIR output for external STEP/DIR driver
				9	*diag1_index* (only with SD_MODE=1) 1: Enable DIAG1 active on index position (microstep look up table position 0)
				10	*diag1_onstate* (only with SD_MODE=1) 1: Enable DIAG1 active when chopper is on (for the coil which is in the second half of the fullstep)
				11	*diag1_steps_skipped* (only with SD_MODE=1) 1: Enable output toggle when steps are skipped in DcStep mode (increment of *LOST_STEPS*). Do not enable in conjunction with other DIAG1 options.

GENERAL CONFIGURATION REGISTERS (0X00…0X0F)					
R/W	Addr	n	Register	Description / bit names	
				12	*diag0_int_pushpull* 0: SWN_DIAG0 is open collector output (active low) 1: Enable SWN_DIAG0 push pull output (active high)
				13	*diag1_poscomp_pushpull* 0: SWP_DIAG1 is open collector output (active low) 1: Enable SWP_DIAG1 push pull output (active high)
				14	*small_hysteresis* 0: Hysteresis for step frequency comparison is 1/16 1: Hysteresis for step frequency comparison is 1/32
				15	*stop_enable* 0: Normal operation 1: Emergency stop: ENCA_DCIN stops the sequencer when tied high (no steps become executed by the sequencer, motor goes to standstill state).
				16	*direct_mode* 0: Normal operation 1: Motor coil currents and polarity directly programmed via serial interface: Register *XTARGET* (0x2D) specifies signed coil A current (bits 8..0) and coil B current (bits 24..16). In this mode, the current is scaled by *IHOLD* setting. Velocity based current regulation of StealthChop is not available in this mode. The automatic StealthChop current regulation will work only for low stepper motor velocities.
				17	*test_mode* 0: Normal operation 1: Enable analog test output on pin ENCN_DCO. *IHOLD*[1..0] selects the function of ENCN_DCO: 0…2: T120, DAC, VDDH *Hint:* Not for user, set to 0 for normal operation!
R+ WC	0x01	3	*GSTAT*	**Bit**	***GSTAT* – Global status flags** (Re-Write with '1' bit to clear respective flags)
				0	*reset* 1: Indicates that the IC has been reset. All registers have been cleared to reset values.
				1	*drv_err* 1: Indicates, that the driver has been shut down due to overtemperature or short circuit detection. Read DRV_STATUS for details. The flag can only be cleared when the temperature is below the limit again.
				2	*uv_cp* 1: Indicates an undervoltage on the charge pump. The driver is disabled during undervoltage. This flag is latched for information.
R	0x02	8	*IFCNT*	Interface transmission counter. This register becomes incremented with each successful UART interface write access. It can be read out to check the serial transmission for lost data. Read accesses do not change the content. Disabled in SPI operation. The counter wraps around from 255 to 0.	

GENERAL CONFIGURATION REGISTERS (0X00…0X0F)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	
W	0x03	8 + 4	*SLAVECONF*	**Bit**	***SLAVECONF***
				7..0	*SLAVEADDR*: These eight bits set the address of unit for the UART interface. The address becomes incremented by one when the external address pin NEXTADDR is active. Range: 0-253 (254 cannot be incremented), *default*=0
				11..8	*SENDDELAY*: 0, 1: 8 bit times (not allowed with multiple slaves) 2, 3: 3*8 bit times 4, 5: 5*8 bit times 6, 7: 7*8 bit times 8, 9: 9*8 bit times 10, 11: 11*8 bit times 12, 13: 13*8 bit times 14, 15: 15*8 bit times
R	0x04	8 + 8	*IOIN*	**Bit**	***INPUT***
					Reads the state of all input pins available
				0	REFL_STEP
				1	REFR_DIR
				2	ENCB_DCEN_CFG4
				3	ENCA_DCIN_CFG5
				4	DRV_ENN
				5	ENC_N_DCO_CFG6
				6	SD_MODE (1=External step and dir source)
				7	SWCOMP_IN (Shows voltage difference of SWN and SWP. Bring DIAG outputs to high level with pushpull disabled to test the comparator.)
				31..24	*VERSION*: 0x30=first version of the IC Identical numbers mean full digital compatibility.
W	0x04	1	*OUTPUT*	**Bit**	***OUTPUT***
					Sets the IO output pin polarity in UART mode
				0	In UART mode, SDO_CFG0 is an output. This bit programs the output polarity of this pin. Its main purpose it to use SDO_CFG0 as NAO next address output signal for chain addressing of multiple ICs. *Hint:* Reset Value is 1 for use as NAO to next IC in single wire chain
W	0x05	32	*X_COMPARE*	Position comparison register for motion controller position strobe. The Position pulse is available on output SWP_DIAG1. *XACTUAL* = *X_COMPARE*: - Output signal PP (position pulse) becomes high. It returns to a low state if the positions mismatch.	
W	0x06		*OTP_PROG*	**Bit**	***OTP_PROGRAM* – OTP programming** Write access programs OTP memory (one bit at a time), Read access refreshes read data from OTP after a write
				2..0	*OTPBIT* Selection of OTP bit to be programmed to the selected byte location (n=0..7: programs bit n to a logic 1)
				5..4	*OTPBYTE* Set to 00

GENERAL CONFIGURATION REGISTERS (0X00…0X0F)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	
				15..8	*OTPMAGIC* Set to *0xbd* to enable programming. A programming time of minimum 10ms per bit is recommended (check by reading *OTP_READ*).
R	0x07		*OTP_READ*	**Bit**	***OTP_READ*** (Access to OTP memory result and update) *See separate table!*
				7..0	*OTP0* byte 0 read data
RW	0x08	5	*FACTORY_ CONF*	4..0	*FCLKTRIM* (*Reset default: OTP*) 0…31: Lowest to highest clock frequency. Check at charge pump output. The frequency span is not guaranteed, but it is tested, that tuning to 12MHz internal clock is possible. The devices come preset to 12MHz clock frequency by OTP programming. *(Reset Default: OTP)*
W	0x09	19	*SHORT_ CONF*	**Bit**	***SHORT_CONF***
				3..0	*S2VS_LEVEL*: Short to VS detector level for lowside FETs. Checks for voltage drop in LS MOSFET and sense resistor. 4 (highest sensitivity) … 15 (lowest sensitivity) *Hint:* Settings from 1 to 3 will trigger during normal operation due to voltage drop on sense resistor. *(Reset Default: OTP 6 or 12)*
				11..8	*S2G_LEVEL*: Short to GND detector level for highside FETs. Checks for voltage drop on high side MOSFET 2 (highest sensitivity) … 15 (lowest sensitivity) *Attention: Settings below 6 not recommended at >52V operation – false detection might result* *(Reset Default: OTP 6 or 12)*
				17..16	*SHORTFILTER*: Spike filtering bandwidth for short detection 0 (lowest, 100ns), 1 (1µs), 2 (2µs) 3 (3µs) *Hint:* A good PCB layout will allow using setting 0. Increase value, if erroneous short detection occurs. *(Reset Default = %01)*
				18	*shortdelay*: Short detection delay 0=750ns: normal, 1=1500ns: high The short detection delay shall cover the bridge switching time. 0 will work for most applications. *(Reset Default = 0)*
W	0x0A	22	*DRV_CONF*	**Bit**	**DRV_CONF**
				4..0	*BBMTIME:* Break-Before make delay 0=shortest (100ns) … 16 (200ns) … 24=longest (375ns) >24 not recommended, use *BBMCLKS* instead *Hint:* Choose the lowest setting safely covering the switching event to avoid bridge cross-conduction. Add roughly 30% of reserve. *(Reset Default = 0)*

GENERAL CONFIGURATION REGISTERS (0X00…0X0F)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	
				11..8	*BBMCLKS*: 0..15: Digital BBM time in clock cycles (typ. 83ns). The longer setting rules (*BBMTIME* vs. *BBMCLKS*). *(Reset Default: OTP 4 or 2)*
				17..16	*OTSELECT*: Selection of over temperature level for bridge disable, switch on after cool down to 120°C / OTPW level. 00: 150°C 01: 143°C 10: 136°C (not recommended when VSA > 24V) 11: 120°C (not recommended, no hysteresis) *Hint:* Adapt overtemperature threshold as required to protect the MOSFETs or other components on the PCB. *(Reset Default = %00)*
				19..18	*DRVSTRENGTH*: Selection of gate driver current. Adapts the gate driver current to the gate charge of the external MOSFETs. 00: weak 01: weak+TC (medium above OTPW level) 10: medium 11: strong *Hint:* Choose the lowest setting giving slopes <100ns. *(Reset Default = %00)*
				21..20	*FILT_ISENSE*: Filter time constant of sense amplifier to suppress ringing and coupling from second coil operation 00: low – 100ns 01: – 200ns 10: – 300ns 11: high– 400ns *Hint:* Increase setting if motor chopper noise occurs due to cross-coupling of both coils. *(Reset Default = %00)*
W	0x0B	8	*GLOBAL SCALER*	7..0	Global scaling of Motor current. This value is multiplied to the current scaling to adapt a drive to a certain motor type. This value should be chosen before tuning other settings because it also influences chopper hysteresis. 0: Full Scale (or write 256) 1 … 31: Not allowed for operation 32 … 255: 32/256 … 255/256 of maximum current. *Hint:* Values >128 recommended for best results *(Reset Default = 0)*
R	0x0C	16	*OFFSET_ READ*	15..8	Offset calibration result phase A (signed)
				7..0	Offset calibration result phase B (signed)

6.1.1 *OTP_READ* – OTP configuration memory

The OTP memory holds power up defaults for certain registers. All OTP memory bits are cleared to 0 by default. Programming only can set bits, clearing bits is not possible. Factory tuning of the clock frequency affects *otp0.0* to *otp0.4*. The state of these bits therefore may differ between individual ICs.

0x07: *OTP_READ* – OTP memory map			
Bit	**Name**	**Function**	**Comment**
7	*otp0.7*	*otp_TBL*	Reset default for *TBL*: 0: *TBL*=%10 (~3µs) 1: *TBL*=%01 (~2µs)
6	*otp0.6*	*otp_BBM*	Reset default for *DRVCONF.BBMCLKS* 0: *BBMCLKS*=4 1: *BBMCLKS*=2
5	*otp0.5*	*otp_S2_LEVEL*	Reset default for *short-detection Levels*: 0: *S2G_LEVEL = S2VS_LEVEL = 6* 1: *S2G_LEVEL = S2VS_LEVEL = 12*
4	*otp0.4*	*OTP_FCLKTRIM*	Reset default for *FCLKTRIM* 0: lowest frequency setting 31: highest frequency setting *Attention: This value is pre-programmed by factory clock trimming to the default clock frequency of 12MHz and differs between individual ICs! It should not be altered.*
3	*otp0.3*		
2	*otp0.2*		
1	*otp0.1*		
0	*otp0.0*		

6.2 Velocity Dependent Driver Feature Control Register Set

VELOCITY DEPENDENT DRIVER FEATURE CONTROL REGISTER SET (0X10...0X1F)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	
				Bit	***IHOLD_IRUN* – Driver current control**
W	0x10	5 + 5 + 4	*IHOLD_IRUN*	4..0	*IHOLD* Standstill current (0=1/32…31=32/32) In combination with StealthChop mode, setting *IHOLD*=0 allows to choose freewheeling or coil short circuit for motor stand still.
				12..8	*IRUN* Motor run current (0=1/32…31=32/32) *Hint:* Choose sense resistors in a way, that normal IRUN is 16 to 31 for best microstep performance.
				19..16	*IHOLDDELAY* Controls the number of clock cycles for motor power down after a motion as soon as standstill is detected (*stst*=1) and *TPOWERDOWN* has expired. The smooth transition avoids a motor jerk upon power down. 0: instant power down 1..15: Delay per current reduction step in multiple of 2^18 clocks
W	0x11	8	*TPOWER DOWN*	*TPOWERDOWN* sets the delay time after stand still (*stst*) of the motor to motor current power down. Time range is about 0 to 4 seconds. *Attention: A minimum setting of 2 is required to allow automatic tuning of StealthChop PWM_OFS_AUTO.* *Reset Default = 10* *0…((2^8)-1)* * 2^18 t_{CLK}	
R	0x12	20	*TSTEP*	Actual measured time between two 1/256 microsteps derived from the step input frequency in units of 1/fCLK. Measured value is (2^20)-1 in case of overflow or stand still. All *TSTEP* related thresholds use a hysteresis of 1/16 of the compare value to compensate for jitter in the clock or the step frequency. The flag *small_hysteresis* modifies the hysteresis to a smaller value of 1/32. (*Txxx**15/16)-1 or (*Txxx**31/32)-1 is used as a second compare value for each comparison value. This means, that the lower switching velocity equals the calculated setting, but the upper switching velocity is higher as defined by the hysteresis setting. When working with the motion controller, the measured *TSTEP* for a given velocity V is in the range (2^{24} / V) ≤ *TSTEP* ≤ 2^{24} / V - 1. In DcStep mode *TSTEP* will not show the mean velocity of the motor, but the velocities for each microstep, which may not be stable and thus does not represent the real motor velocity in case it runs slower than the target velocity.	

<table>
<tr><th colspan="5">VELOCITY DEPENDENT DRIVER FEATURE CONTROL REGISTER SET (0X10…0X1F)</th></tr>
<tr><th>R/W</th><th>Addr</th><th>n</th><th>Register</th><th>Description / bit names</th></tr>
<tr><td>W</td><td>0x13</td><td>20</td><td>TPWMTHRS</td><td>This is the upper velocity for StealthChop voltage PWM mode.
TSTEP ≥ TPWMTHRS
- StealthChop PWM mode is enabled, if configured
- DcStep is disabled</td></tr>
<tr><td>W</td><td>0x14</td><td>20</td><td>TCOOLTHRS</td><td>This is the lower threshold velocity for switching on smart energy CoolStep and StallGuard feature. (unsigned)

Set this parameter to disable CoolStep at low speeds, where it cannot work reliably. The stop on stall function (enable with sg_stop when using internal motion controller) and the stall output signal become enabled when exceeding this velocity. In non-DcStep mode, it becomes disabled again once the velocity falls below this threshold.

TCOOLTHRS ≥ TSTEP ≥ THIGH:
- CoolStep is enabled, if configured
- StealthChop voltage PWM mode is disabled

TCOOLTHRS ≥ TSTEP
- Stop on stall is enabled, if configured
- Stall output signal (DIAG0/1) is enabled, if configured</td></tr>
<tr><td>W</td><td>0x15</td><td>20</td><td>THIGH</td><td>This velocity setting allows velocity dependent switching into a different chopper mode and fullstepping to maximize torque. (unsigned)
The stall detection feature becomes switched off for 2-3 electrical periods whenever passing THIGH threshold to compensate for the effect of switching modes.

TSTEP ≤ THIGH:
- CoolStep is disabled (motor runs with normal current scale)
- StealthChop voltage PWM mode is disabled
- If vhighchm is set, the chopper switches to chm=1 with TFD=0 (constant off time with slow decay, only).
- If vhighfs is set, the motor operates in fullstep mode, and the stall detection becomes switched over to DcStep stall detection.</td></tr>
</table>

Microstep velocity time reference t for velocities: TSTEP = f_{CLK} / $f_{STEP256}$. *TSTEP* is related to 1/256 microstep resolution independent of actual resolution set by *MRES*.

6.3 Ramp Generator Registers

6.3.1 Ramp Generator Motion Control Register Set

RAMP GENERATOR MOTION CONTROL REGISTER SET (0X20…0X2D)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	**Range [Unit]**
RW	0x20	2	*RAMPMODE*	*RAMPMODE:* 0: Positioning mode (using all A, D and V parameters) 1: Velocity mode to positive *VMAX* (using *AMAX* acceleration) 2: Velocity mode to negative *VMAX* (using *AMAX* acceleration) 3: Hold mode (velocity remains unchanged, unless stop event occurs)	0…3
RW	0x21	32	*XACTUAL*	Actual motor position (signed) *Hint:* This value normally should only be modified, when homing the drive. In positioning mode, modifying the register content will start a motion.	-2^31… +(2^31)-1
R	0x22	24	*VACTUAL*	Actual motor velocity from ramp generator (signed) The sign matches the motion direction. A negative sign means motion to lower *XACTUAL*.	+-(2^23)-1 [µsteps / t]
W	0x23	18	*VSTART*	Motor start velocity (unsigned) For universal use, set *VSTOP* ≥ *VSTART.* This is not required if the motion distance is sufficient to ensure deceleration from *VSTART* to *VSTOP*.	0…(2^18)-1 [µsteps / t]
W	0x24	16	*A1*	First acceleration between *VSTART* and *V1* (unsigned)	0…(2^16)-1 [µsteps / ta²]
W	0x25	20	*V1*	First acceleration / deceleration phase threshold velocity (unsigned) 0: Disables *A1* and *D1* phase, use *AMAX*, *DMAX* only	0…(2^20)-1 [µsteps / t]
W	0x26	16	*AMAX*	Second acceleration between *V1* and *VMAX* (unsigned) This is the acceleration and deceleration value for velocity mode.	0…(2^16)-1 [µsteps / ta²]
W	0x27	23	*VMAX*	Motion ramp target velocity (for positioning ensure *VMAX* ≥ *VSTART*) (unsigned) This is the target velocity in velocity mode. It can be changed any time during a motion.	0…(2^23)-512 [µsteps / t]
W	0x28	16	*DMAX*	Deceleration between *VMAX* and *V1* (unsigned)	0…(2^16)-1 [µsteps / ta²]
W	0x2A	16	*D1*	Deceleration between *V1* and *VSTOP* (unsigned) *Attention: Do not set 0 in positioning mode, even if V1=0!*	1…(2^16)-1 [µsteps / ta²]

RAMP GENERATOR MOTION CONTROL REGISTER SET (0X20…0X2D)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	**Range [Unit]**
W	0x2B	18	*VSTOP*	Motor stop velocity (unsigned) *Hint:* Set VSTOP ≥ VSTART to allow positioning for short distances *Attention: Do not set 0 in positioning mode, minimum 10 recommend, set >100 for faster ramp termination!*	1…(2^18)-1 [µsteps / t] Reset Default=1
W	0x2C	16	*TZEROWAIT*	Defines the waiting time after ramping down to zero velocity before next movement or direction inversion can start. Time range is about 0 to 2 seconds. This setting avoids excess acceleration e.g. from *VSTOP* to *-VSTART*.	0…(2^16)-1 * 512 t_{CLK}
RW	0x2D	32	*XTARGET*	Target position for ramp mode (signed). Write a new target position to this register in order to activate the ramp generator positioning in *RAMPMODE*=0. Initialize all velocity, acceleration, and deceleration parameters before. *Hint:* The position is allowed to wrap around, thus, *XTARGET* value optionally can be treated as an unsigned number. *Hint:* The maximum possible displacement is +/-((2^31)-1). *Hint:* When increasing *V1, D1 or DMAX during* a motion, rewrite *XTARGET* afterwards to trigger a second acceleration phase, if desired.	-2^31… +(2^31)-1

6.3.2 Ramp Generator Driver Feature Control Register Set

RAMP GENERATOR DRIVER FEATURE CONTROL REGISTER SET (0X30…0X36)				
R/W	**Addr**	**n**	***Register***	**Description / *bit names***
W	0x33	23	*VDCMIN*	Automatic commutation DcStep becomes enabled above velocity *VDCMIN* (unsigned) (only when using internal ramp generator, not for STEP/DIR interface – in STEP/DIR mode, DcStep becomes enabled by the external signal DCEN) In this mode, the actual position is determined by the sensor-less motor commutation and becomes fed back to *XACTUAL*. In case the motor becomes heavily loaded, *VDCMIN* also is used as the minimum step velocity. Activate stop on stall (*sg_stop*) to detect step loss. 0: Disable, DcStep off *|VACT|* ≥ *VDCMIN* ≥ 256: - Triggers the same actions as exceeding *THIGH* setting. - Switches on automatic commutation DcStep *Hint:* Also set *DCCTRL* parameters to operate DcStep. (Only bits 22… 8 are used for value and for comparison)
RW	0x34	12	***SW_MODE***	Switch mode configuration *See separate table!*
R+ WC	0x35	14	***RAMP_STAT***	Ramp status and switch event status *See separate table!*
R	0x36	32	*XLATCH*	Ramp generator latch position, latches *XACTUAL* upon a programmable switch event (see *SW_MODE*). *Hint:* The encoder position can be latched to *ENC_LATCH* together with *XLATCH* to allow consistency checks.

Time reference t for velocities: $t = 2^{24} / f_{CLK}$
Time reference ta² for accelerations: $ta^2 = 2^{41} / (f_{CLK})^2$

6.3.2.1 SW_MODE – Reference Switch & StallGuard2 Event Configuration Register

0X34: *SW_MODE* – REFERENCE SWITCH AND STALLGUARD2 EVENT CONFIGURATION REGISTER		
Bit	**Name**	**Comment**
11	*en_softstop*	0: Hard stop 1: Soft stop The soft stop mode always uses the deceleration ramp settings *DMAX*, V1, *D1*, *VSTOP* and *TZEROWAIT* for stopping the motor. A stop occurs when the velocity sign matches the reference switch position (REFL for negative velocities, REFR for positive velocities) and the respective switch stop function is enabled. A hard stop also uses *TZEROWAIT* before the motor becomes released. *Attention: Do not use soft stop in combination with StallGuard2. Use soft stop for StealthChop operation <u>at high velocity.</u> In this case, hard stop must be avoided, as it could result in severe overcurrent.*
10	*sg_stop*	1: Enable stop by StallGuard2 (also available in DcStep mode). Disable to release motor after stop event. Program *TCOOLTHRS* for velocity threshold. *Hint:* Do not enable during motor spin-up, wait until the motor velocity exceeds a certain value, where StallGuard2 delivers a stable result. This velocity threshold should be programmed using TCOOLTHRS.
9	*en_latch_encoder*	1: Latch encoder position to *ENC_LATCH* upon reference switch event.
8	*latch_r_inactive*	1: Activates latching of the position to *XLATCH* upon an inactive going edge on the right reference switch input REFR. The active level is defined by *pol_stop_r*.
7	*latch_r_active*	1: Activates latching of the position to *XLATCH* upon an active going edge on the right reference switch input REFR. *Hint:* Activate *latch_r_active* to detect any spurious stop event by reading *status_latch_r*.
6	*latch_l_inactive*	1: Activates latching of the position to *XLATCH* upon an inactive going edge on the left reference switch input REFL. The active level is defined by *pol_stop_l*.
5	*latch_l_active*	1: Activates latching of the position to *XLATCH* upon an active going edge on the left reference switch input REFL. *Hint:* Activate *latch_l_active* to detect any spurious stop event by reading *status_latch_l*.
4	*swap_lr*	1: Swap the left and the right reference switch input REFL and REFR
3	*pol_stop_r*	Sets the active polarity of the right reference switch input 0=non-inverted, high active: a high level on REFR stops the motor 1=inverted, low active: a low level on REFR stops the motor
2	*pol_stop_l*	Sets the active polarity of the left reference switch input 0=non-inverted, high active: a high level on REFL stops the motor 1=inverted, low active: a low level on REFL stops the motor
1	*stop_r_enable*	1: Enables automatic motor stop during active right reference switch input *Hint:* The motor restarts in case the stop switch becomes released.
0	*stop_l_enable*	1: Enables automatic motor stop during active left reference switch input *Hint:* The motor restarts in case the stop switch becomes released.

6.3.2.2 *RAMP_STAT – Ramp & Reference Switch Status Register*

0X35: *RAMP_STAT* – RAMP AND REFERENCE SWITCH STATUS REGISTER			
R/W	**Bit**	**Name**	**Comment**
R	13	*status_sg*	1: Signals an active StallGuard2 input from the CoolStep driver or from the DcStep unit, if enabled. *Hint:* When polling this flag, stall events may be missed – activate *sg_stop* to be sure not to miss the stall event.
R+ WC	12	*second_move*	1: Signals that the automatic ramp required moving back in the opposite direction, e.g., due to on-the-fly parameter change (Write '1' to clear)
R	11	*t_zerowait_ active*	1: Signals, that *TZEROWAIT* is active after a motor stop. During this time, the motor is in standstill.
R	10	*vzero*	1: Signals, that the actual velocity is 0.
R	9	*position_ reached*	1: Signals, that the target position is reached. This flag becomes set while *XACTUAL* and *XTARGET* match.
R	8	*velocity_ reached*	1: Signals, that the target velocity is reached. This flag becomes set while *VACTUAL* and *VMAX* match.
R+ WC	7	*event_pos_ reached*	1: Signals, that the target position has been reached (*position_reached* becoming active). (Write '1' to clear flag and interrupt condition) This bit is ORed to the *interrupt output* signal.
R+ WC	6	*event_stop_ sg*	1: Signals an active StallGuard2 stop event. Resetting the register will clear the stall condition and the motor may re-start motion unless the motion controller has been stopped. (Write '1' to clear flag and interrupt condition) This bit is ORed to the *interrupt output* signal.
R	5	*event_stop_r*	1: Signals an active stop right condition due to stop switch. The stop condition and the interrupt condition can be removed by setting *RAMP_MODE* to hold mode or by commanding a move to the opposite direction. In *soft_stop* mode, the condition will remain active until the motor has stopped motion into the direction of the stop switch. Disabling the stop switch or the stop function also clears the flag, but the motor will continue motion. This bit is ORed to the *interrupt output* signal.
	4	*event_stop_l*	1: Signals an active stop left condition due to stop switch. The stop condition and the interrupt condition can be removed by setting *RAMP_MODE* to hold mode or by commanding a move to the opposite direction. In *soft_stop* mode, the condition will remain active until the motor has stopped motion into the direction of the stop switch. Disabling the stop switch or the stop function also clears the flag, but the motor will continue motion. This bit is ORed to the *interrupt output* signal.
R+ WC	3	*status_latch_r*	1: Latch right ready (enable position latching using *SW_MODE* settings *latch_r_active* or *latch_r_inactive*) (Write '1' to clear)
	2	*status_latch_l*	1: Latch left ready (enable position latching using *SW_MODE* settings *latch_l_active* or *latch_l_inactive*) (Write '1' to clear)
R	1	*status_stop_r*	Reference switch right status (1=active)
	0	*status_stop_l*	Reference switch left status (1=active)

6.4 Encoder Registers

Attention:
The encoder interface is not available in Step&Direction mode, as the encoder pins serve a different function in that mode.

ENCODER REGISTER SET (0X38…0X3C)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	**Range [Unit]**
RW	0x38	11	***ENCMODE***	Encoder configuration and use of N channel *See separate table!*	
RW	0x39	32	*X_ENC*	Actual encoder position (signed)	-2^31… +(2^31)-1
W	0x3A	32	*ENC_CONST*	Accumulation constant (signed) 16 bit integer part, 16 bit fractional part *X_ENC* accumulates +/- *ENC_CONST* / (2^16**X_ENC*) (binary) or +/-*ENC_CONST* / (10^4**X_ENC*) (decimal) *ENCMODE* bit *enc_sel_decimal* switches between decimal and binary setting. Use the sign, to match rotation direction!	binary: ± [µsteps/2^16] ±(0 … 32767.999847) decimal: ±(0.0 … 32767.9999) *reset default = 1.0 (=65536)*
R+ WC	0x3B	2	*ENC_STATUS*	Encoder status information bit 0: *n_event* bit 1*: deviation_warn* 1: Event detected. To clear the status bit, write with a 1 bit at the corresponding position. Deviation_warn cannot be cleared while a warning still persists. Set *ENC_DEVIATION* zero to disable. Both bits are ORed to the *interrupt output* signal.	
R	0x3C	32	*ENC_LATCH*	Encoder position *X_ENC* latched on N event	
W	0x3D	20	*ENC_ DEVIATION*	Maximum number of steps deviation between encoder counter and XACTUAL for deviation warning Result in flag ENC_STATUS.*deviation_warn* 0=Function is off.	

6.4.1 *ENCMODE* – Encoder Register

0x38: *ENCMODE* – Encoder register			
Bit	**Name**	**Comment**	
10	*enc_sel_decimal*	0	Encoder prescaler divisor binary mode: Counts *ENC_CONST(fractional part)* /65536
		1	Encoder prescaler divisor decimal mode: Counts in *ENC_CONST(fractional part)* /10000
9	*latch_x_act*	1: Also latch *XACTUAL* position together with *X_ENC*. Allows latching the ramp generator position upon an N channel event as selected by *pos_edge* and *neg_edge*.	
8	*clr_enc_x*	0	Upon N event, *X_ENC* becomes latched to *ENC_LATCH* only
		1	Latch and additionally clear encoder counter *X_ENC* at N-event
7	*neg_edge*	***n p***	**N channel event sensitivity**
6	*pos_edge*	0 0	N channel event is active during an active N event level
		0 1	N channel is valid upon active going N event
		1 0	N channel is valid upon inactive going N event
		1 1	N channel is valid upon active going and inactive going N event
5	*clr_once*	1: Latch or latch and clear *X_ENC* on the next N event following the write access	
4	*clr_cont*	1: Always latch or latch and clear *X_ENC* upon an N event (once per revolution, it is recommended to combine this setting with edge sensitive N event)	
3	*ignore_AB*	0	An N event occurs only when polarities given by *pol_N*, *pol_A* and *pol_B* match.
		1	Ignore A and B polarity for N channel event
2	*pol_N*	Defines active polarity of N (0=low active, 1=high active)	
1	*pol_B*	Required B polarity for an N channel event (0=neg., 1=pos.)	
0	*pol_A*	Required A polarity for an N channel event (0=neg., 1=pos.)	

6.5 Motor Driver Registers

MICROSTEPPING CONTROL REGISTER SET (0X60…0X6B)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	**Range [Unit]**
W	0x60	32	*MSLUT[0]* microstep table entries 0…31	Each bit gives the difference between entry x and entry x+1 when combined with the corresponding *MSLUTSEL W* bits: 0: *W*= %00: -1 %01: +0 %10: +1 %11: +2 1: *W*= %00: +0 %01: +1 %10: +2 %11: +3 This is the differential coding for the first quarter of a wave. Start values for *CUR_A* and *CUR_B* are stored for *MSCNT* position 0 in *START_SIN* and *START_SIN90*. *ofs31, ofs30, …, ofs01, ofs00* … *ofs255, ofs254, …, ofs225, ofs224*	32x 0 or 1 *reset default= sine wave table*
W	0x61 … 0x67	7 x 32	*MSLUT[1...7]* microstep table entries 32…255		7x 32x 0 or 1 *reset default= sine wave table*
W	0x68	32	***MSLUTSEL***	This register defines four segments within each quarter *MSLUT* wave. Four 2-bit entries determine the meaning of a 0 and a 1 bit in the corresponding segment of *MSLUT*. *See separate table!*	*0<X1<X2<X3* *reset default= sine wave table*
W	0x69	8 + 8	*MSLUTSTART*	bit 7… 0: *START_SIN* bit 23… 16: *START_SIN90* *START_SIN* gives the absolute current at microstep table entry 0. *START_SIN90* gives the absolute current for microstep table entry at positions 256. Start values are transferred to the microstep registers *CUR_A* and *CUR_B*, whenever the reference position *MSCNT*=0 is passed.	*START_SIN* *reset default =0* *START_SIN90* *reset default =247*
R	0x6A	10	*MSCNT*	Microstep counter. Indicates actual position in the microstep table for *CUR_B*. *CUR_A* uses an offset of 256 (2 phase motor). *Hint:* Move to a position where *MSCNT* is zero before re-initializing *MSLUTSTART* or *MSLUT* and *MSLUTSEL*.	0…1023
R	0x6B	9 + 9	*MSCURACT*	bit 8… 0: *CUR_B* (signed): Actual microstep current for motor phase B (sine wave) as read from MSLUT (not scaled by current) bit 24… 16: *CUR_A* (signed): Actual microstep current for motor phase A (co-sine wave) as read from *MSLUT* (not scaled by current)	+/-0...255

DRIVER REGISTER SET (0X6C…0X7F)					
R/W	**Addr**	**n**	***Register***	**Description / *bit names***	**Range [Unit]**
RW	0x6C	32	***CHOPCONF***	chopper and driver configuration *See separate table!*	*reset default=* *0x10410150*
W	0x6D	25	***COOLCONF***	CoolStep smart current control register and StallGuard2 configuration *See separate table!*	
W	0x6E	24	*DCCTRL*	DcStep (*DC*) automatic commutation configuration register (enable via pin DCEN or via *VDCMIN*): bit 9… 0: *DC_TIME*: Upper PWM on time limit for commutation (*DC_TIME* * 1/f_{CLK}). Set slightly above effective blank time *TBL*. bit 23… 16: *DC_SG*: Max. PWM on time for step loss detection using DcStep StallGuard2 in DcStep mode. (*DC_SG* * 16/f_{CLK}) Set slightly higher than *DC_TIME*/16 0=disable *Hint:* Using a higher microstep resolution or interpolated operation, DcStep delivers a better StallGuard signal. DC_SG is also available above VHIGH if vhighfs is activated. For best result also set vhighchm.	
R	0x6F	32	***DRV_ STATUS***	StallGuard2 value and driver error flags *See separate table!*	
W	0x70	32	***PWMCONF***	Voltage PWM mode chopper configuration *See separate table!*	*reset default=* *0xC40C001E*
R	0x71	9+8	*PWM_SCALE*	Results of StealthChop amplitude regulator. These values can be used to monitor automatic PWM amplitude scaling (255=max. voltage).	
				bit 7… 0: *PWM_SCALE_SUM*: Actual PWM duty cycle. This value is used for scaling the values *CUR_A* and *CUR_B* read from the sine wave table.	0…255
				bit 24… 16: *PWM_SCALE_AUTO*: 9 Bit signed offset added to the calculated PWM duty cycle. This is the result of the automatic amplitude regulation based on current measurement.	signed -255…+255
R	0x72	8+8	*PWM_AUTO*	These automatically generated values can be read out in order to determine a default / power up setting for *PWM_GRAD* and *PWM_OFS*.	
				bit 7… 0: *PWM_OFS_AUTO*: Automatically determined offset value	0…255

DRIVER REGISTER SET (0X6C…0X7F)						
R/W	**Addr**	**n**	***Register***	**Description / *bit names***		**Range [Unit]**
				bit 23… 16	*PWM_GRAD_AUTO*: Automatically determined gradient value	0…255
R	0x73	20	*LOST_STEPS*	Number of input steps skipped due to higher load in DcStep operation, if step input does not stop when DC_OUT is low. This counter wraps around after 2^20 steps. Counts up or down depending on direction. Only with SDMODE=1.		

MICROSTEP TABLE CALCULATION FOR A SINE WAVE EQUIVALENT TO THE POWER ON DEFAULT

$$round\left(248 * sin\left(2 * PI * \frac{i}{1024} + \frac{PI}{1024}\right)\right) - 1$$

- *i*:[0… 255] is the table index
- The amplitude of the wave is 248. The resulting maximum positive value is 247 and the maximum negative value is -248.
- The round function rounds values from 0.5 to 1.4999 to 1

6.5.1 *MSLUTSEL* – Look up Table Segmentation Definition

<table>
<tr><th colspan="4">0X68: MSLUTSEL – LOOK UP TABLE SEGMENTATION DEFINITION</th></tr>
<tr><th>Bit</th><th>Name</th><th>Function</th><th>Comment</th></tr>
<tr><td>31</td><td rowspan="8"><i>X3</i></td><td rowspan="8">LUT segment 3 start</td><td rowspan="24">The sine wave look-up table can be divided into up to four segments using an individual step width control entry <i>Wx</i>. The segment borders are selected by <i>X1</i>, <i>X2</i> and <i>X3</i>.

Segment 0 goes from 0 to <i>X1</i>-1.
Segment 1 goes from <i>X1</i> to <i>X2</i>-1.
Segment 2 goes from <i>X2</i> to <i>X3</i>-1.
Segment 3 goes from <i>X3</i> to 255.

For defined response the values shall satisfy:
0<<i>X1</i><<i>X2</i><<i>X3</i></td></tr>
<tr><td>30</td></tr>
<tr><td>29</td></tr>
<tr><td>28</td></tr>
<tr><td>27</td></tr>
<tr><td>26</td></tr>
<tr><td>25</td></tr>
<tr><td>24</td></tr>
<tr><td>23</td><td rowspan="8">X2</td><td rowspan="8">LUT segment 2 start</td></tr>
<tr><td>22</td></tr>
<tr><td>21</td></tr>
<tr><td>20</td></tr>
<tr><td>19</td></tr>
<tr><td>18</td></tr>
<tr><td>17</td></tr>
<tr><td>16</td></tr>
<tr><td>15</td><td rowspan="8">X1</td><td rowspan="8">LUT segment 1 start</td></tr>
<tr><td>14</td></tr>
<tr><td>13</td></tr>
<tr><td>12</td></tr>
<tr><td>11</td></tr>
<tr><td>10</td></tr>
<tr><td>9</td></tr>
<tr><td>8</td></tr>
<tr><td>7</td><td rowspan="2">W3</td><td rowspan="2">LUT width select from ofs(X3) to ofs255</td><td rowspan="8">Width control bit coding W0…W3:
%00: MSLUT entry 0, 1 select: -1, +0
%01: MSLUT entry 0, 1 select: +0, +1
%10: MSLUT entry 0, 1 select: +1, +2
%11: MSLUT entry 0, 1 select: +2, +3</td></tr>
<tr><td>6</td></tr>
<tr><td>5</td><td rowspan="2">W2</td><td rowspan="2">LUT width select from ofs(X2) to ofs(X3-1)</td></tr>
<tr><td>4</td></tr>
<tr><td>3</td><td rowspan="2">W1</td><td rowspan="2">LUT width select from ofs(X1) to ofs(X2-1)</td></tr>
<tr><td>2</td></tr>
<tr><td>1</td><td rowspan="2">W0</td><td rowspan="2">LUT width select from ofs00 to ofs(X1-1)</td></tr>
<tr><td>0</td></tr>
</table>

6.5.2 *CHOPCONF* – Chopper Configuration

0X6C: *CHOPCONF* – CHOPPER CONFIGURATION			
Bit	**Name**	**Function**	**Comment**
31	*diss2vs*	short to supply protection disable	0: Short to VS protection is on 1: Short to VS protection is disabled
30	*diss2g*	short to GND protection disable	0: Short to GND protection is on 1: Short to GND protection is disabled
29	*dedge*	enable double edge step pulses	1: Enable step impulse at each step edge to reduce step frequency requirement.
28	*intpol*	interpolation to 256 microsteps	1: The actual microstep resolution (*MRES*) becomes extrapolated to 256 microsteps for smoothest motor operation (useful for STEP/DIR operation, only)
27	*mres3*	*MRES* micro step resolution	%0000: Native 256 microstep setting. Normally use this setting with the internal motion controller.
26	*mres2*		
25	*mres1*		
24	*mres0*		%0001 … %1000: 128, 64, 32, 16, 8, 4, 2, FULLSTEP Reduced microstep resolution esp. for STEP/DIR operation. The resolution gives the number of microstep entries per sine quarter wave. The driver automatically uses microstep positions which result in a symmetrical wave, when choosing a lower microstep resolution. step width=2^*MRES* [microsteps]
23	*tpfd3*	*TPFD* passive fast decay time	*TPFD* allows dampening of motor mid-range resonances. Passive fast decay time setting controls duration of the fast decay phase inserted after bridge polarity change N_{CLK}= 128**TPFD* %0000: Disable %0001 … %1111: 1 … 15
22	*tpfd2*		
21	*tpdf1*		
20	*tpfd0*		
19	*vhighchm*	high velocity chopper mode	This bit enables switching to *chm*=1 and *fd*=0, when *VHIGH* is exceeded. This way, a higher velocity can be achieved. Can be combined with *vhighfs*=1. If set, the *TOFF* setting automatically becomes doubled during high velocity operation in order to avoid doubling of the chopper frequency.
18	*vhighfs*	high velocity fullstep selection	This bit enables switching to fullstep, when *VHIGH* is exceeded. Switching takes place only at 45° position. The fullstep target current uses the current value from the microstep table at the 45° position.
17	-	reserved	reserved, set to 0
16	*tbl1*	*TBL* blank time select	%00 … %11: Set comparator blank time to 16, 24, 36 or 54 clocks *Hint*: %01 or %10 is recommended for most applications *(Reset Default: OTP %01 or %10)*
15	*tbl0*		
14	*chm*	chopper mode	0: Standard mode (SpreadCycle) 1: Constant off time with fast decay time. Fast decay time is also terminated when the negative nominal current is reached. Fast decay is after on time.
13	-	reserved	Reserved, set to 0
12	*disfdcc*	fast decay mode	*chm*=1: *disfdcc*=1 disables current comparator usage for termination of the fast decay cycle

0x6C: *CHOPCONF* – Chopper Configuration				
Bit	**Name**	**Function**	**Comment**	
11	*fd3*	*TFD [3]*	*chm*=1: MSB of fast decay time setting *TFD*	
10	*hend3*	*HEND* hysteresis low value *OFFSET* sine wave offset	*chm*=0	%0000 … %1111: Hysteresis is -3, -2, -1, 0, 1, …, 12 (1/512 of this setting adds to current setting) This is the hysteresis value which becomes used for the hysteresis chopper.
9	*hend2*		*chm*=1	%0000 … %1111: Offset is -3, -2, -1, 0, 1, …, 12 This is the sine wave offset and 1/512 of the value becomes added to the absolute value of each sine wave entry.
8	*hend1*			
7	*hend0*			
6	*hstrt2*	*HSTRT* hysteresis start value added to *HEND*	*chm*=0	%000 … %111: Add 1, 2, …, 8 to hysteresis low value *HEND* (1/512 of this setting adds to current setting) *Attention: Effective HEND+HSTRT ≤ 16.* *Hint:* Hysteresis decrement is done each 16 clocks
5	*hstrt1*	*TFD [2..0]* fast decay time setting	*chm*=1	Fast decay time setting (MSB: *fd3*): %0000 … %1111: Fast decay time setting *TFD* with N_{CLK}= 32**TFD* (%0000: slow decay only)
4	*hstrt0*			
3	*toff3*	*TOFF* off time and driver enable	Off time setting controls duration of slow decay phase N_{CLK}= 24 + 32**TOFF* %0000: Driver disable, all bridges off %0001: 1 – use only with *TBL* ≥ 2 %0010 … %1111: 2 … 15	
2	*toff2*			
1	*toff1*			
0	*toff0*			

6.5.3 *COOLCONF* – Smart Energy Control CoolStep and StallGuard2

0x6D: *COOLCONF* – Smart Energy control CoolStep and StallGuard2			
Bit	**Name**	**Function**	**Comment**
…	-	reserved	set to 0
24	*sfilt*	StallGuard2 filter enable	0: Standard mode, high time resolution for StallGuard2 1: Filtered mode, StallGuard2 signal updated for each four fullsteps (resp. six fullsteps for 3 phase motor) only to compensate for motor pole tolerances
23	-	reserved	set to 0
22	*sgt6*	StallGuard2 threshold value	This signed value controls StallGuard2 level for stall output and sets the optimum measurement range for readout. A lower value gives a higher sensitivity. Zero is the starting value working with most motors. -64 to +63: A higher value makes StallGuard2 less sensitive and requires more torque to indicate a stall.
21	*sgt5*		
20	*sgt4*		
19	*sgt3*		
18	*sgt2*		
17	*sgt1*		
16	*sgt0*		
15	*seimin*	minimum current for smart current control	0: 1/2 of current setting (*IRUN*) 1: 1/4 of current setting (*IRUN*)
14	*sedn1*	current down step speed	%00: For each 32 StallGuard2 values decrease by one %01: For each 8 StallGuard2 values decrease by one %10: For each 2 StallGuard2 values decrease by one %11: For each StallGuard2 value decrease by one
13	*sedn0*		
12	-	reserved	set to 0
11	*semax3*	StallGuard2 hysteresis value for smart current control	If the StallGuard2 result is equal to or above (*SEMIN*+*SEMAX*+1)*32, the motor current becomes decreased to save energy. %0000 … %1111: 0 … 15
10	*semax2*		
9	*semax1*		
8	*semax0*		
7	-	reserved	set to 0
6	*seup1*	current up step width	Current increment steps per measured StallGuard2 value %00 … %11: 1, 2, 4, 8
5	*seup0*		
4	-	reserved	set to 0
3	*semin3*	minimum StallGuard2 value for smart current control and smart current enable	If the StallGuard2 result falls below *SEMIN**32, the motor current becomes increased to reduce motor load angle. %0000: smart current control CoolStep off %0001 … %1111: 1 … 15
2	*semin2*		
1	*semin1*		
0	*semin0*		

6.5.4 *PWMCONF* – Voltage PWM Mode StealthChop

0x70: *PWMCONF* – VOLTAGE MODE PWM STEALTHCHOP			
Bit	**Name**	**Function**	**Comment**
31 30 29 28	*PWM_LIM*	PWM automatic scale amplitude limit when switching on	Limit for *PWM_SCALE_AUTO* when switching back from SpreadCycle to StealthChop. This value defines the upper limit for bits 7 to 4 of the automatic current control when switching back. It can be set to reduce the current jerk during mode change back to StealthChop. It does not limit *PWM_GRAD* or *PWM_GRAD_AUTO* offset. (Default = 12)
27 26 25 24	*PWM_REG*	Regulation loop gradient	User defined maximum PWM amplitude change per half wave when using *pwm_autoscale*=1. (1…15): 1: 0.5 increments (slowest regulation) 2: 1 increment 3: 1.5 increments 4: 2 increments *(Reset default)*) … 8: 4 increments ... 15: 7.5 increments (fastest regulation)
23	-	reserved	set to 0
22	-	reserved	set to 0
21 20	*freewheel1* *freewheel0*	Allows different standstill modes	Stand still option when motor current setting is zero (*I_HOLD*=0). %00: Normal operation %01: Freewheeling %10: Coil shorted using LS drivers %11: Coil shorted using HS drivers
19	*pwm_autograd*	PWM automatic gradient adaptation	0: Fixed value for *PWM_GRAD* (*PWM_GRAD_AUTO* = *PWM_GRAD*) 1: Automatic tuning (only with *pwm_autoscale*=1) (*Reset default*) *PWM_GRAD_AUTO* is initialized with *PWM_GRAD* while *pwm_autograd*=0 and becomes optimized automatically during motion. <u>Preconditions</u> 1. *PWM_OFS_AUTO* has been automatically initialized. This requires standstill at *IRUN* for >130ms to a) detect standstill b) wait > 128 chopper cycles at *IRUN* and c) regulate *PWM_OFS_AUTO* so that -1 < *PWM_SCALE_AUTO* < 1 2. Motor running and *PWM_SCALE_SUM* < 255 and 1.5 * *PWM_OFS_AUTO* * (*IRUN*+1)/32 < *PWM_SCALE_SUM* < 4* *PWM_OFS_AUTO* * (*IRUN*+1)/32. <u>Time required for tuning *PWM_GRAD_AUTO*</u> About 8 fullsteps per change of +/-1. Also enables use of reduced chopper frequency for tuning *PWM_OFS_AUTO*.

0X70: *PWMCONF* – VOLTAGE MODE PWM STEALTHCHOP			
Bit	**Name**	**Function**	**Comment**
18	*pwm_ autoscale*	PWM automatic amplitude scaling	0: User defined feed forward PWM amplitude. The current settings *IRUN* and *IHOLD* are not enforced by regulation, but scale the PWM amplitude, only! The resulting PWM amplitude (limited to 0…255) is: *PWM_OFS * ((CS_ACTUAL+1) / 32)* *+ PWM_GRAD * 256 / TSTEP* 1: Enable automatic current control *(Reset default)*
17	*pwm_freq1*	PWM frequency selection	%00: f_{PWM}=2/1024 f_{CLK} *(Reset default)* %01: f_{PWM}=2/683 f_{CLK} %10: f_{PWM}=2/512 f_{CLK} %11: f_{PWM}=2/410 f_{CLK}
16	*pwm_freq0*		
15	*PWM_ GRAD*	User defined amplitude gradient	Velocity dependent gradient for PWM amplitude: *PWM_GRAD * 256 / TSTEP* This value is added to *PWM_OFS* to compensate for the velocity-dependent motor back-EMF. Use *PWM_GRAD* as initial value for automatic scaling to speed up the automatic tuning process. To do this, set *PWM_GRAD* to the determined, application specific value, with *pwm_autoscale*=0. Only afterwards, set *pwm_autoscale*=1. Enable StealthChop when finished. *Hint:* After initial tuning, the required initial value can be read out from *PWM_GRAD_AUTO.*
14			
13			
12			
11			
10			
9			
8			
7	*PWM_ OFS*	User defined amplitude (offset)	User defined PWM amplitude offset (0-255) related to full motor current (*CS_ACTUAL*=31) in stand still. (*Reset default*=30) Use *PWM_OFS* as initial value for automatic scaling to speed up the automatic tuning process. To do this, set *PWM_OFS* to the determined, application specific value, with *pwm_autoscale*=0. Only afterwards, set *pwm_autoscale*=1. Enable StealthChop when finished. *PWM_OFS* = 0 will disable scaling down motor current below a motor specific lower measurement threshold. This setting should only be used under certain conditions, i.e., when the power supply voltage can vary up and down by a factor of two or more. It prevents the motor going out of regulation, but it also prevents power down below the regulation limit. *PWM_OFS* > 0 allows automatic scaling to low PWM duty cycles even below the lower regulation threshold. This allows low (standstill) current settings based on the actual (hold) current scale (register *IHOLD_IRUN*).
6			
5			
4			
3			
2			
1			
0			

6.5.5 *DRV_STATUS* – StallGuard2 Value and Driver Error Flags

0X6F: DRV_STATUS – STALLGUARD2 VALUE AND DRIVER ERROR FLAGS			
Bit	**Name**	**Function**	**Comment**
31	*stst*	standstill indicator	This flag indicates motor stand still in each operation mode. This occurs 2^20 clocks after the last step pulse.
30	*olb*	open load indicator phase B	1: Open load detected on phase A or B. *Hint:* This is just an informative flag. The driver takes no action upon it. False detection may occur in fast motion and standstill. Check during slow motion, only.
29	*ola*	open load indicator phase A	
28	*s2gb*	short to ground indicator phase B	1: Short to GND detected on phase A or B. The driver becomes disabled. The flags stay active, until the driver is disabled by software (*TOFF*=0) or by the DRV_ENN input.
27	*s2ga*	short to ground indicator phase A	
26	*otpw*	overtemperature pre-warning flag	1: Overtemperature pre-warning threshold is exceeded. The overtemperature pre-warning flag is common for both bridges.
25	*ot*	overtemperature flag	1: Overtemperature limit has been reached. Drivers become disabled until *otpw* is also cleared due to cooling down of the IC. The overtemperature flag is common for both bridges.
24	*StallGuard*	StallGuard2 status	1: Motor stall detected (*SG_RESULT*=0) or DcStep stall in DcStep mode.
23 22 21	-	reserved	Ignore these bits
20 19 18 17 16	*CS ACTUAL*	actual motor current / smart energy current	Actual current control scaling, for monitoring smart energy current scaling controlled via settings in register *COOLCONF*, or for monitoring the function of the automatic current scaling.
15	*fsactive*	full step active indicator	1: Indicates that the driver has switched to fullstep as defined by chopper mode settings and velocity thresholds.
14	*stealth*	StealthChop indicator	1: Driver operates in StealthChop mode
13	*s2vsb*	short to supply indicator phase B	1: Short to supply detected on phase A or B. The driver becomes disabled. The flags stay active, until the driver is disabled by software (*TOFF*=0) or by the DRV_ENN input. Sense resistor voltage drop is included in the measurement!
12	*s2vsa*	short to supply indicator phase A	
11	-	reserved	Ignore this bit
10	-	reserved	Ignore this bit
9 8 7 6 5 4 3 2 1 0	*SG_ RESULT*	StallGuard2 result respectively PWM on time for coil A in standstill for motor temperature detection	Mechanical load measurement: The StallGuard2 result gives a means to measure mechanical motor load. A higher value means lower mechanical load. A value of 0 signals highest load. With optimum *SGT* setting, this is an indicator for a motor stall. The stall detection compares *SG_RESULT* to 0 to detect a stall. *SG_RESULT* is used as a base for CoolStep operation, by comparing it to a programmable upper and a lower limit. It is not applicable in StealthChop mode. StallGuard2 works best with microstep operation or DcStep. Temperature measurement: In standstill, no StallGuard2 result can be obtained. *SG_RESULT* shows the chopper on-time for motor coil A instead. Move the motor to a determined microstep position at a certain current setting to get a rough estimation of motor temperature by a reading the chopper on-time. As the motor heats up, its coil resistance rises and the chopper on-time increases.